Category: Cybersecurity

PS-036: PQC DNSSEC MTL Benchmark Using a Reference Implementation

- **Description:**
  - Classical public-key algorithms used in DNSSEC are vulnerable to future quantum cryptanalysis. Post-Quantum Cryptography (PQC) signatures resist quantum attacks, but their large public keys and signatures cause DNS UDP responses to exceed standard network packet limits, causing fragmentation and dropped lookups. The IETF Merkle Tree Ladder (MTL) mode (draft-harvey-cfrg-mtl-mode) uses compact authentication paths to reduce transmitted signature overhead. In this project, you will benchmark a supplied reference MTL implementation for one supported PQC algorithm across three dataset sizes, comparing signing latency, verification time, and transmitted byte volume against equivalent workloads without MTL.
- **Challenge:**
  - Benchmark a supplied Merkle Tree Ladder (MTL) mode implementation for one supported post-quantum cryptography (PQC) signature algorithm under repeatable DNSSEC-related workloads. Compare equivalent workloads with and without MTL where supported.
- **Key Requirements**
  - Use 1 supported PQC signature algorithm.
  - Use 1 mentor-tested MTL reference implementation.
  - Test 3 dataset sizes.
  - Compare equivalent workloads with and without MTL where supported.
  - Run repeated trials under a pinned software and hardware environment.
  - Measure signing time, verification time and transmitted bytes.
  - Record and report benchmark configuration and workload parameters.
  - Separate directly measured cache usage from analytical estimates.
  - Defer the project if the required implementation and test fixtures are not available before launch.
  - No new cryptography, resolver implementation from scratch, Internet deployment or claimed security proof.
- **Expected Outcome**
  - Pinned benchmark environment
  - Benchmark harness
  - MTL and non-MTL workload configurations
  - Three dataset-size test cases
  - Repeated-trial measurements
  - Signing and verification performance results
  - Transmitted-byte comparison
  - Cache-use measurements and analytical estimates
  - Reproducible results report
  - Technical demonstration

- ## Deliverables*

1. GitHub Collaboration: (**This step is crucial for allowing mentors to provide real-time oversight.**)
    - Each team must set up a private repository on GitHub.
    - Add our GitHub account (https://github.com/aiori-hackathon) as a collaborator.
    - Once the collaboration request has been accepted all the concerned users (all the teammates as well as aiori-hackathon account) transfer the ownership of the repository to our GitHub account. (https://github.com/aiori-hackathon)
    - Ensure that your repository is named according to your team's name.
2. Each team is required to upload a PDF to their GitHub repository
    - Please follow the specified structure: Proposed-structure-hackathon.pdf
3. Prototype Demonstration with a Pseudocode Snippet to the mentor

- ## References
    - IETF Draft | PQC DNSSEC MTL Mode
    - IETF 126 Hackathon